

PEICHI HUANG

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Education

University of Michigan, Ann Arbor

Master of Data Science - GPA: 4.0/4.0

- Information Retrieval (A+), Machine Learning (A), Data Mining (A), Web Systems, Parallel Programming with GPUs.

Aug. 2024 – Present

Michigan, United State

National Tsing Hua University

B.S, Mathematics and Computer Science - Major-GPA: 3.84/4.0 (Academic Excellence Award)

Sep. 2019 – Jan. 2024

Hsinchu City, Taiwan

Technical Skills

Programming Languages: C, C++, Python, Java, JavaScript, HTML, CSS, SQL, R, Kotlin

Developer Tools: Google Cloud Platform, Colaboratory, Git, PyTorch, TensorFlow, Scikit-Learn, Keras, Spark

Experience

School of Information, University of Michigan

Graduate Student Instructor for SI650/EECS549 Information Retrieval

Aug. 2025 – Dec. 2025

Michigan, US

- Led weekly discussion sections and held office hours to support students in understanding concepts in information retrieval and NLP, providing guidance on assignments and projects.
- Coordinated with the instructional team to manage course delivery, grading, and logistics, ensuring efficient communication and smooth execution of course operations.

BlockChain Security Corporation

R&D Intern

Jun. 2025 – Aug. 2025

Taipei City, Taiwan

- Constructed an OCR pipeline for identifying recovery seed phrases in real-world images by combining image processing, text recognition using PaddleOCR, and phrase validation using a Trie-based matching over the BIP-39 dictionary.
- Built a lightweight filtering system using Laplacian blur detection to improve efficiency and increase recall rate by 40%.

Information Technology Service Center, NYCU

Research Assistant

Feb. 2024 – Aug. 2025

Taipei City, Taiwan

- Evaluated the AI chatbot's performance on the center's website through NLP methods, and analyzed the chatbot usage.
- Performed exploratory data analysis (EDA) on post-graduate year medical student training outcome data and trained ML models such as regression and random forest with more than 0.85 accuracy to provide insight on future training.

Biomedica Corporation

R&D AI Engineer Intern

Mar. 2024 – Aug. 2024

New Taipei City, Taiwan

- Trained deep learning models with YOLO-v8 and YOLO-v9 frameworks to automate image segmentation, achieving an average MAP of over 0.90 and over 0.95 with image augmentation.
- Developed algorithms for image quality assessment, and disease diagnosis via segmentation with over 0.85 accuracy.

Institute of Information Science, Academia Sinica

Software Engineer Intern

Jul. 2022 – Jan. 2024

Taipei City, Taiwan

- Engineered a digital diary application using Android Studio, enabling the collection of clinical data from over 200 patients diagnosed with GERD while integrating an online questionnaire for comprehensive health reporting.

Publications

A Retrieval-based Chatbot for Customer Service Using Large Language Model

Pei-Chi Huang (Best Presenter Award), Mei-Lien Pan

IEEE ICFTSS, 2024

Kuala Lumpur, Malaysia

Effect of Training Content and Environment Factors on Resident Retention Rates

Hsiao-Chin Shen, Pei-Chi Huang, Mei-Lien Pan, Ching-Weng Yang, et al.

JCMA, 2025

Taipei, Taiwan, 2025

Projects

SAIT, Study Abroad Information for Taiwanese Students | Vue, Git, Python, ML

Fall 2024

- Built a similarity-based admission analysis tool to recommend relevant study abroad cases using weighted embedding.

GUI-ASFormer, Detecting Keyframes in GUI Videos | Python, ML, DL

Winter 2025

- Improved ASFormer for GUI action segmentation with a two stream encoder, high-resolution embeddings, frame differencing, and multiscale temporal convolutions, boosting F1@50 by +7.1 and edit distance by +6.8.

Job Postings Searching System | Python, Information Retrieval

Fall 2024

- Implemented a system that used user profiles as queries to match relevant job postings, applying NLP techniques such as text classification, semantic similarity, and ranking models to achieve a 67% improvement in NDCG over baseline.